Demonstrating Aleatoric Uncertainty in Remaining Useful Life Prediction Using LSTM with Probabilistic Layer



Ahmad Kamal Bin Mohd Nor, Srinivasa Rao Pedapati, Masdi Muhammad, and Mohd Amin Abdul Majid

Abstract A Remaining Useful Life prediction with Aleatoric uncertainty is presented in this paper. A Long Short-Term Memory (LSTM) architecture with probabilistic layer is employed where a normal distribution layer is incorporated to produce the predicted Health Index (HI) distribution of turbofan engines. Compared to the performance of other point estimates techniques in the literature, the probabilistic LSTM achieved a competitive performance in predicting the turbofan's RUL and RUL sequence and have the advantage to express the level of uncertainty along its sequence prediction. This work is important as it reflect a real-world deep learning application where uncertainty indication is needed to evaluate prediction for important decision-making process.

Keywords Aleatoric uncertainty · Probabilistic neural network · Remaining useful life · CMAPPS · Turbofan

1 Introduction

Safety and human lives protection are an utmost priority in the aerospace industry. Sudden failure of aerospace assets during operational mode is the worst-case scenario that demand full attention from industry players and reliability practitioner. To this end, Prognostic and Health Management (PHM) has emerged as one of the key domains in reducing major accidents, enhancing asset's reliability, and improving availability while at the same time optimizing maintenance cost [1, 2].

PHM activities consist of fault detection and isolation, fault diagnostic and prognostic, i.e., predicting time of failure, as well as health management [3]. In the prognostic field, the remaining useful life (RUL) prediction, which estimates the equipment's remaining time before failure is particularly challenging [4].

Currently, 3 categories of RUL prediction techniques are widely employed: model-based, data-driven based and the hybrid of both methods [1, 3, 4]. While

A. K. Bin Mohd Nor · S. R. Pedapati (✉) · M. Muhammad · M. A. Abdul Majid
Mechanical Engineering Department, Universiti Teknologi PETRONAS, Perak 32610 Seri Iskandar, Malaysia
e-mail: srinivasa.pedapati@utp.edu.my

© Institute of Technology PETRONAS Sdn Bhd 2023

F. Ahmad et al. (eds.), *ICPER 2020*, Lecture Notes in Mechanical Engineering,
https://doi.org/10.1007/978-981-19-1939-8_41

being more accurate in prediction, model-based methods, which models the underlying physic properties of the system, are difficult to develop. Data-driven methods however thrive with the ever-increasing amount of data coupled with intelligent sensor and data-processing power. Amongst all, deep-learning (DL) has proven to be the most powerful in the last few years [2]. This technique essentially consists of training models with historical run-to-failure data, making it to learn the nonlinear relationships from the data in a black-box fashion [3].

As a black-box approach, the prediction of DL models is not easily interpretable. Most of the available work in the literature were developed to predict scalar point estimates without any account of uncertainty. Naturally, this hinder its application in the real-world, safety-sensitive, high investment domains where uncertainty of the estimation is vital before any costly decision-making. As an example, according to our founding, only 4 out of 282 publications [5–8] linked to the famous CMAPPS Turbofan dataset from 2008 to this day include the presentation of prediction uncertainty. However, none explain in detail the interpretation of uncertainty obtained. Literature review is done on the publications presented in a CMAPPS dataset systematic review paper [9] (106 papers) together with additional publication until July 2020 (176 papers) using the methodology presented in [9] on 9 libraries: IEEE Xplore, ScienceDirect, Springer, AIS, Wiley Online Library, Taylor & Francis, Sage Publishing, Emerald Insight and Scopus.

1.1 Uncertainty in Deep Learning.

Uncertainty in deep learning linked to the quality of input data is known as Aleatoric uncertainties [10, 11, 13–15]. This uncertainty can happen due to noise, data acquisition error or stochasticity captured in the input data, which is the usual situation encountered in the real world [10, 12–15]. This type of uncertainty cannot be reduced further by having more data if no improvement is done on the data acquisition method.

This paper analyzes and interpret the Aleatoric uncertainty of turbofan engine RUL predictions using LSTM neural network with probabilistic layer. It puts into light the randomness in the output prediction linked to deep learning that need to be considered especially for vital application's decision-making process.

2 Related Works

Uncertainty quantification work in deep learning is currently being used in many different fields, catering for specific application. Bayesian deep learning (BDL) is incorporated in convolution neural network (CNN), and Bidirectional LSTM in [6] to predict the remaining useful life of bearing with uncertainty. In [16] a probabilistic normalized convolutional neural network (NCNNs) is used on a depth

completion dataset. A prognostic method with uncertainty quantification involving hydraulic mechanisms of circuit breakers using BDL coupled with a sequential Bayesian boosting algorithm is presented in [17]. In [18], a diagnostic of Diabetic retinopathy by DR|Graduate, a deep learning-based DR grading CAD system framework with uncertainty is proposed. A probabilistic Super-Resolved q-Space deep learning method is applied in [19] to measure brain tissue microstructure. Monte Carlo dropout procedure is executed in [20] to provide confidence in neural network leaf skeleton photo reconstruction. A Bayesian deep multisource learning (BDMSL) is used to diagnose glaucoma disease with uncertainty in [21]. In [22], a BDL with Variational Inference and Monte Carlo method is proposed to study the influence of inflow uncertainty on reservoir operation rules in the Three Gorges Dam of the Yangtze River. Input-dependent noise model and variational dropout is used in [23] to provide uncertainty in medical image enhancement problem. Bayesian inference is used to predict plant problem with uncertainty in [24].

3 Methodology

3.1 LSTM Architecture with Probabilistic Layer.

Regression is done using LSTM architecture whose optimum layers and hyperparameters obtained via Bayesian hyperparameter optimization [25]. 500 evaluations are done to find the best hyperparameters in the search space presented in Table 1 that minimize the validation error. The training target are HI corresponding to the normalized value of the RUL target for each cycle. A Normal distribution layer is incorporated as the output layer of the LSTM network, outputting both mean and standard deviation of the HI prediction.

LSTM architecture is increasingly used in time-series data modelling due to the capability of learning long-term dependencies [9]. LSTM employs gating mechanism to regulate the data flow and retain important memory or forget invaluable ones [26]. LSTM consists of Input Layer, Hidden Layer and Output Layer with a recurrent mechanism enabling the output of the Hidden Layer to be added with new input. This enable the modeling of sequential data.

Table 1 BayesOpt search space

Layers	LSTM hidden units	Dense layer size	Dropout rate	Mini batch size	Max epoch	Learning rate
Search space						
1 LSTM 1 Dense 1 Dropout	10 to 1000	10 to 500	0.000 to 0.500	26 to 130	10 to 2000	5e-4 to 1e-3

LSTM gating equation:

$$f_t = \sigma\left(W_f.X_t + R_f.h_{t-1} + b_f\right) \tag{1}$$

$$i_t = \sigma(W_i.X_t + R_i.h_{t-1} + b_i) \tag{2}$$

$$o_t = \sigma(W_o.X_t + R_o.h_{t-1} + b_o) \tag{3}$$

$$M_t = \varphi(W_c.X_t + R_c.h_{t-1} + b_c) \tag{4}$$

$$C_t = f_t * C_{t-1} + i_t * M_t \tag{5}$$

$$h_t = o_t * \varnothing(C_t) \tag{6}$$

W* = Input Weight

R* = Recurrent Weight

b* = bias

$\sigma, \varphi, \varnothing$ = activationfunctions

Equations (1) to (3) are Forget Gate, Input Gate and Output Gate equations. Equation (4) is used to calculate the candidate value that will be added to the new state "t" (5) together with the old state "t–1", regulated by the Forget Gate. (6) represents the final output of LSTM. "." is matrix multiplication while "" is point to point multiplication [27].

3.2 Network Performance Assessment.

We assess the performance of our network by calculating the Root Mean Squared Error (RMSE) and Scoring Function of the obtained RUL prediction as respectively shown in Eqs. (7) and (8), (9) and (10) [28]. The scoring function gives higher score for the same error in early prediction than late prediction. It penalizes late prediction than the early ones.

$$RMSE = \sqrt{\frac{1}{M}\sum_{i=1}^{M}\left(RUL_{truth}^{(i)} - RUL_{predicted}^{(i)}\right)^2} \tag{7}$$

$$s = M\sum_{i=1}^{M} s_i \tag{8}$$

$$s = \left\{ \begin{array}{l} e^{\frac{-d_i}{13}} - 1, d_i < 0 \\ e^{\frac{di}{10}} - 1, d_i > 0 \end{array} \right\} \tag{9}$$

$$di = \left(RUL_{predicted} - RUL_{true}\right) \tag{10}$$

3.3 *Uncertainty Quantification of RUL and of HI Sequence Prediction*

The LSTM will predict a sequence of HI normal distributions time series for each fleet. Then we sample 1000 points and calculate the mean,$\mu_{(i)}$ and standard deviation, $\sigma_{(i)}$ of each distribution, obtaining a sequence of HI distribution mean for each fleet. We obtain the RUL prediction by taking the first HI distribution mean in the sequence and reverse the normalization. We plot the predicted HI distribution mean sequence together with standard deviations as Upper and Lower bounds as shown in Eqs. (11) and (12), to visualize the area of uncertainty. The Aleatoric Uncertainty is represented by the distance between the predicted HI distribution mean curve and the Upper and Lower bounds. To have more detail about the uncertainty of the prediction, we plot the standard deviations versus time and observe their variation behavior.

$$Upper\ Bound_{Fleet(i)} = \mu_{(i)} + 2\sigma_{(i)} \tag{11}$$

$$Lower\ Bound_{Fleet(i)} = \mu_{(i)} - 2\sigma_{(i)} \tag{12}$$

4 Case Study

CMAPPS ((Commercial Modular Aero Propulsion System Simulation) Turbofan runto-failure datasets were published in 2008 by Nasa Prognostic Centre (PCoE) of Ames 2 Research Centre (http://ti.arc.nasa.gov/projects/dataprognostics) consisting of 4 complete sets of Training, Testing and ground truth RUL for numerous turbofan engines (Fig. 1). The simulated data was obtained by simulating a variety of operational conditions and injecting faults of varying degree degradation.

The chosen training dataset and testing data consist of 260 recorded turbofan degradations. Each Training and Testing data correspond to a turbofan whose health condition deteriorate after certain cycle [29]. Each fleet training and testing data are time series sequence comprising of Time (Cycle), 3 Operating Conditions (OC) and 21 sensors measurements as in Table 2, without any RUL information. The OC refers to different operating regimes combination of Altitude (O-42 K ft.), Throttle

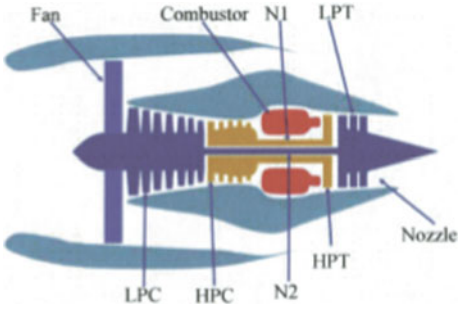


Fig. 1 Turbofan engine diagram [30]

Table 2 Sensors description

Sensor	Ref	Description	Unit
S1	T2	Total temperature fan inlet	^{0}R
S2	T24	Total temperature at LPC outlet	^{0}R
S3	T30	Total temperature at HPC outlet	^{0}R
S4	T50	Total temperature at LPT outlet	^{0}R
S5	P2	Pressure at fan inlet	psia
S6	P15	Total pressure in bypass-duct	psia
S7	P30	Total pressure at HPC outlet	psia
S8	Nf	Physical fan speed	rpm
S9	Nc	Physical core speed	rpm
S10	Epr	Engine pressure ratio (P50/P2)	N/A
S11	Ps30	Static pressure at HPC outlet	psia
S12	Phi	Ratio of fuel flow to Ps30	Pps/psi
S13	NRf	Corrected fan speed	rpm
S14	NRc	Corrected core speed	rpm
S15	BPR	Bypass ratio	N/A
S16	farB	Burner fuel–air ratio	N/A
S17	htBleed	Bleed enthalphy	N/A
S18	Nf_dmd	Demanded fan speed	rpm
S19	PCNfR_dmd	Demanded corrected fan speed	rpm
S20	W31	HPT coolant bleed	lbm/s
S21	W32	LPT coolant bleed	lbm/s

Table 3 Signal trend

Monotonic signal
2,3,4,7,8,9,11,12,13,14,15,17,20,21

Resolver Angle (20–100), and Mach Number (0–0.84) [30]. High levels of noise are incorporated, and the faults encountered are hidden by the effect of various operational conditions [30]. Only the ground truth RUL for each turbofan, are given. We thus need to do data preparation to select the appropriate sensors, visualize the sensor trends and extract each cycle target HI.

4.1 Degradation Assumption

Each fleet might be used in different operating conditions. As such the extend of degradation is different from one another. Sensors information extracted are crucial in addition to flight duration and flight conditions to model the accumulated degradation. The degradation is mainly due to wear and tear due to usage pattern that progress into component faults. The type of component fault could be single or a combination of any of these failures: Fan, Low Pressure Compressor (LPC), High Pressure Compressor (HPC), High Pressure Turbine (HPT) and Low-Pressure Turbine (LPT) (Table 1) [30].

4.2 Data Preparation

4.2.1 Sensor Selection

14 strictly monotonic sensors are selected as presented in Table 3 [28, 31]. Strictly monotonic sensors are useful as they best represent single trending degradation contrary to irregular and unchanged signals. The corresponding sensors are sensors 2,3,4,7,8,9,11,12,13,14,15,17,20,21.

4.2.2 Piece-Wise Linear Degradation

To obtain RUL sequence labels for training, piece-wise linear degradation model is employed [32, 33]. Each fleet health is thus considered stable in the beginning until the start of deterioration which is the failure start point that initiate a linear degradation until failure.

Each fleet is associated with time series corresponding to the total operational duration of the fleet and the last cycle indicate the final instant before failure. Thus initially, we model the RUL of a fleet to start with the value of the last cycle and

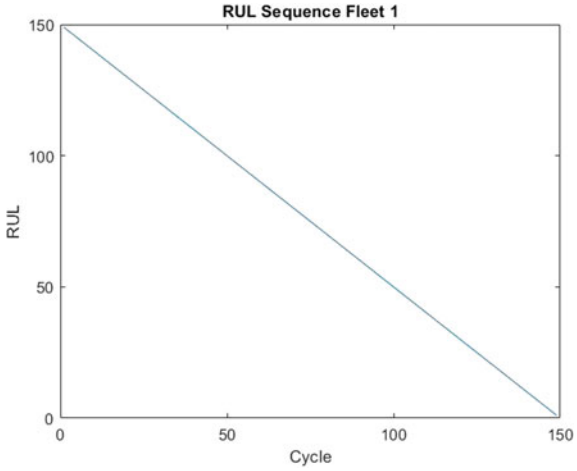


Fig. 2 Initial RUL for Fleet 1

Table 4 Mean of sensor's response means

Sensor	S2	S3	S4	S7	S11	S12	S15	S20	S21	Mean
Cycle	49	51	50	62	44	56	50	65	58	**53.8**

degrade linearly until 0. As seen in Fig. 2, Fleet 1 has a cycle that start from 1 to 149. Considering a linear degradation, Fleet 1 initially has an RUL of 149 that decreases until 0.

The failure start point in each sensor is obtained by using Cumulative Sum (CUSUM) anomaly detection technique, which returns the first index of the upper and lower cumulative sums of each sensor signal that have drifted beyond 5 standard deviations above and below a target mean, indicating the initiating points of degradation [34]. Then, the mean of the failure start points for each sensor can then be calculated. From these means, we can calculate the mean of the means of the sensor's detection, which is our failure start point. The means are presented in Table 4.

Combining the linear degradation obtained earlier and the failure start point, the transformed Fleet 1 RUL sequence is presented in Fig. 3.

4.2.3 Final RUL Quantification Process and HI Establishment

To enable generalization, we set the RUL that are above 100 to 100. We then normalize the new RUL into HI using Eq. (13) that takes the value of 1 (healthy) and 0 (failure). The piece-wise HI sequence curve is shown in Fig. 4 [35].

$$HI_{(t)} = \frac{RUL_{max} - RUL_{(t)}}{RUL_{max} - RUL_{min}} \tag{13}$$

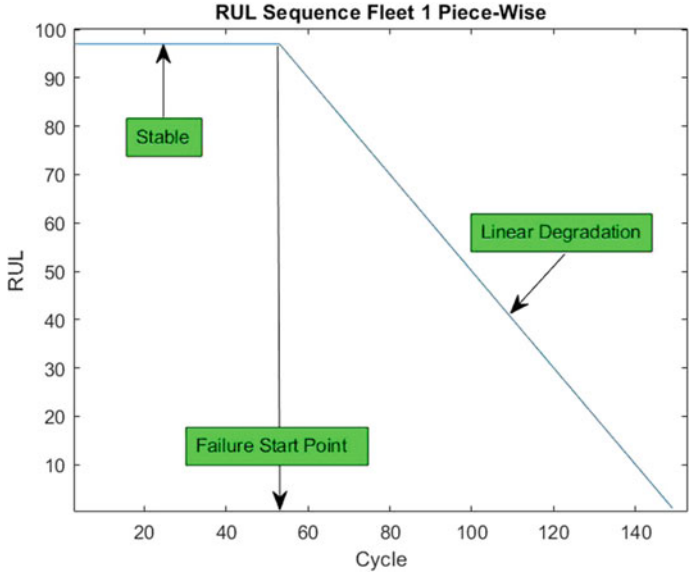


Fig. 3 Final RUL sequence Fleet 1

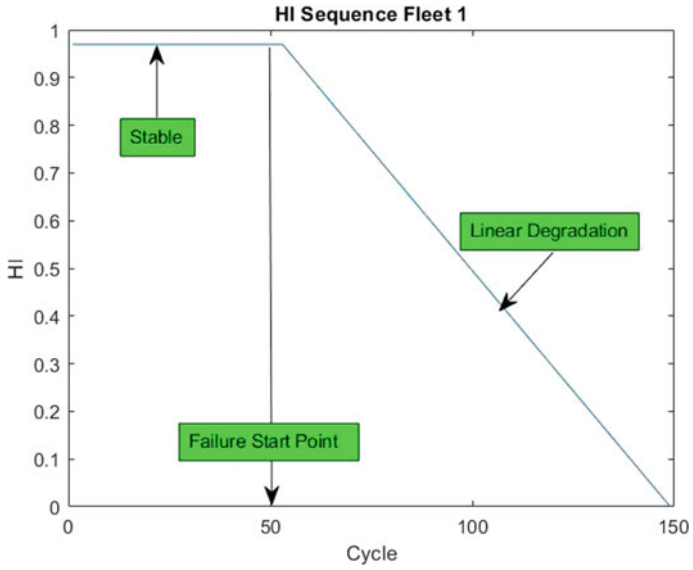


Fig. 4 Final HI Fleet 1

$RUL_{(t)}$ = RUL at time t

RUL_{max} = Health State RUL

RUL_{min} = Final RUL Before Failure

The LSTM training inputs will consist of OCs and the selected sensors while the HI will be the training response. We split randomly 20% of training data as validation data. The testing data consists of OCs and selected sensors from the testing dataset.

5 Result and Discussion

5.1 Network Performance

We obtained lowest validation error with the optimized hyperparameters as listed in Table 5.

The network achieves competitive performance compared to other point estimates data-driven methods in the literature like shown in Table 6. We need to consider that this is just a simple LSTM neural network compared to other more complicated structures such as improved Grey Model (GM) [38] and DeepCNN [37]. The training of the model is also different and more difficult than point estimates methods as the model outputs distributions rather than single point prediction. In term of early failure prediction, the model score is also competitive with early prediction amounting to 67% of the whole fleet of turbofan as shown in Table 7.

Table 5 BayesOpt optimized parameters

Layers	Hidden units	Dense size	Mini batch size	Dropout	Max epoch	Learning rate
1 LSTM 1 Dense 1 Dropout	*Optimized Parameters*					
	635	331	43	0.03	14	5.613e-4

Table 6 RMSE and score of models

Method	RMSE	Score
SVM [36]	51.5	N/A
DNN [36]	41.8	N/A
MLP [37]	37.6	17,972
Proposed Method	**24.1**	**1095**
Improved GM [38]	23.0	1193
DeepCNN [37]	18.5	1287

Table 7 Early RUL prediction versus late RUL prediction

Early	Late
67%	33%

5.2 *Aleatoric Uncertainty for HI Sequence Prediction*

For illustration purpose, we present the HI and HI sequence result of Turbofan 2 and 10. As shown in Figs. 5 and 7, the HI mean, or RUL mean prediction of the Turbofan 2 and 10 are not far from the ground truth. We can also see from Figs. 6 and 8 that the model gives early failure degradation prediction for both fleets.

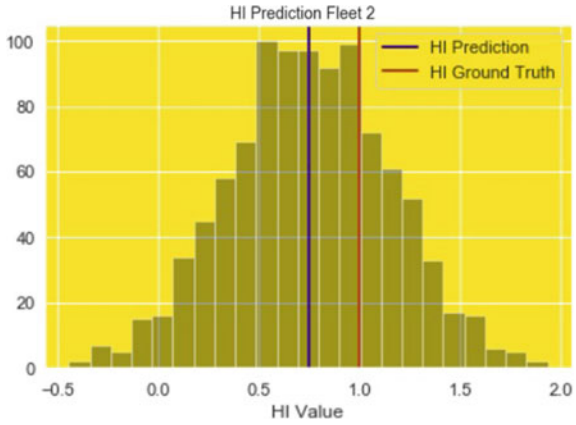


Fig. 5 HI prediction distribution Fleet 2

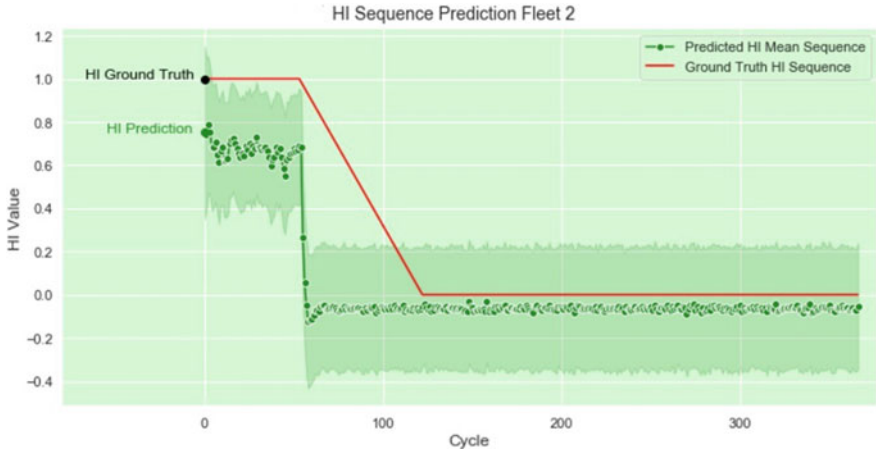


Fig. 6 HI sequence prediction Fleet 2

From the illustrations given in Figs. 6 and 8 we can see that the model gives early HI prediction for Fleet 2 and 10. We can also verify from Figs. 6 and 8 that the predicted HI sequence for these fleets follow the trend of the ground truth HI sequence.

From the standard deviation plots in Figs. 9 and 10, we can see that all cases start with a high uncertainty before decreasing to a low point. It signifies the high uncertainty associated with predicting the initial HI in each case, translated as the imprecise prediction of RUL.

Then, the uncertainty shows monotonic fluctuations for a certain period before increasing rapidly. These oscillations signify that the model predicts with a near-stable level of uncertainty on the true healthy state of the engine which is monotonic.

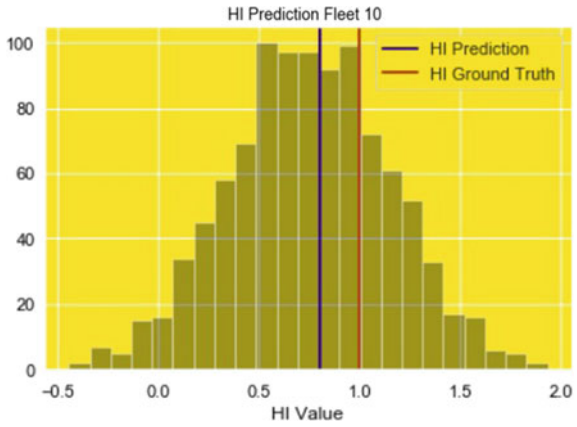


Fig. 7 HI prediction distribution Fleet 10

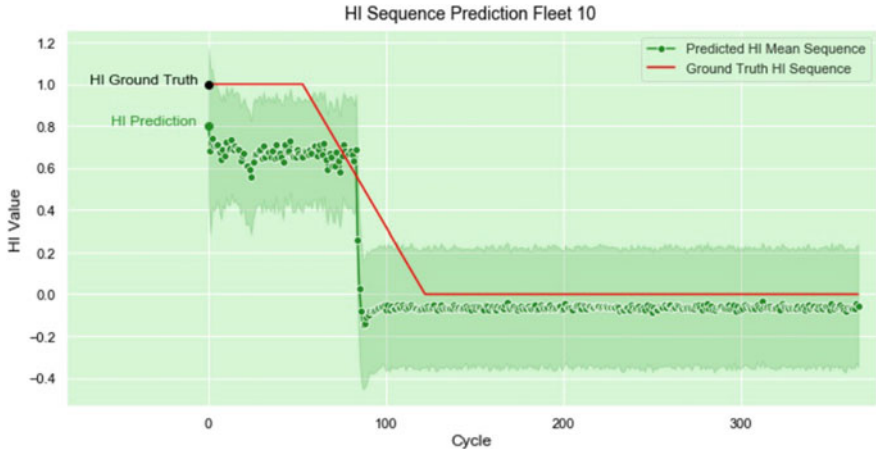


Fig. 8 HI sequence prediction Fleet 10

As shown in Figs. 6 and 8, the predicted sequence seems to follow the ground truth trend, but a gap exists between the predicted HI mean sequence and ground truth sequence, thus the oscillation. Then the sudden increase in uncertainty value indicates that the model predicted the start of deterioration as shown in Figs. 9 and 10. As we can see in Figs. 6 and 8, the HI sequence prediction decrease is steeper than the ground truth curve. Then the uncertainty decreases gradually to a small fluctuation until the end of the cycle as shown in Figs. 6 and 8 as well as 9 and 10. Along the decrease, we see in Figs. 6 and 8 that the model predicted the HI to be 0 earlier than the ground truth, thus explaining the high uncertainty before stabilization as shown in Figs. 9 and 10. Finally, like the healthy state, the model predicted with a near-stable level of uncertainty the end-of-life state of the turbofan, which is also monotonic.

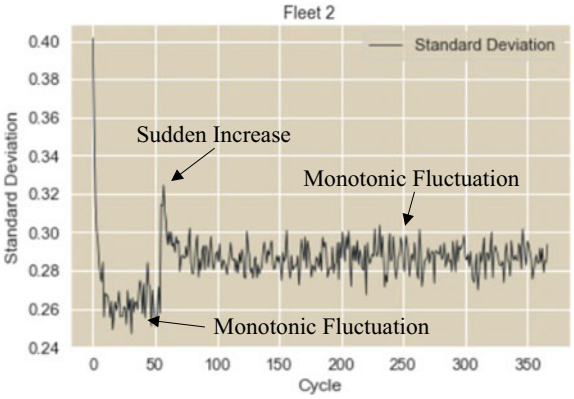


Fig. 9 HI prediction uncertainty Fleet 2

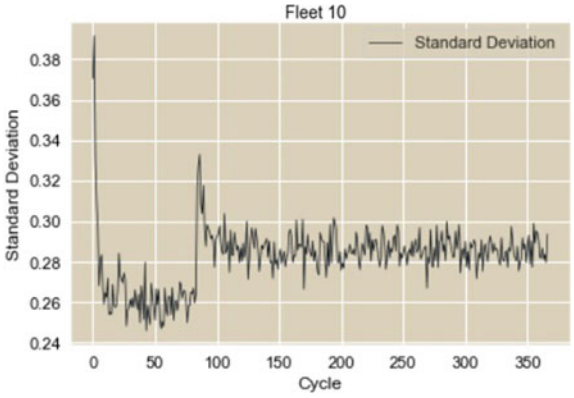


Fig. 10 HI prediction uncertainty Fleet 10

6 Conclusion

In this paper, the Aleatoric uncertainty influenced by noisy sensorial data is put into evidence in an RUL prediction problem using a probabilistic LSTM neural network. Normal distribution layer is incorporated at the output layer of the neural network to produce the predicted Health Index (HI) distribution. The network achieved a competitive performance in predicting the RUL and RUL sequence of turbofan fleets compared to other point estimate methods. By examining the bounded HI sequence prediction plot and the standard deviations, we show and interpret the level of uncertainty along the time series prediction.

Acknowledgements The authors would like to acknowledge Dr. Mohd Amin Abdul Majid and the sponsorship of Newton Link Grant No. 015ME0-015 for this conference paper.

References

1. Si X, Ren Z (2019) A data-fusion based prognostic method for complex degrading system. 2019 Prognostics and system health management conference (PHM-Qingdao), Qingdao, China, pp 1–6. https://doi.org/10.1109/PHM-Qingdao46334.2019.8943070
2. Ellefsen AL, Ushakov S, Æsøy V, Zhang H (2019) Validation of data-driven labeling approaches using a novel deep network structure for remaining useful life predictions. IEEE Access 7:71563–71575. https://doi.org/10.1109/ACCESS.2019.2920297
3. Li F et al (2018)A light gradient boosting machine for Remainning useful life estimation of aircraft engines. 2018 21st international conference on intelligent transportation systems (ITSC), Maui, HI, pp 3562–3567, https://doi.org/10.1109/ITSC.2018.8569801
4. Ge Y, Sun L, Ma J (2019) An Improved PF remaining useful life prediction method based on quantum genetics and LSTM. IEEE Access 7:160241–160247. https://doi.org/10.1109/ACCESS.2019.2951197
5. Kim M, Liu K (2020) A Bayesian deep learning framework for interval estimation of remaining useful life in complex systems by incorporating general degradation characteristics. IISE Trans,https://doi.org/10.1080/24725854.2020.1766729
6. Peng W, Ye Z, Chen N (March 2020) Bayesian deep-learning-based health prognostics toward prognostics uncertainty. IEEE Trans Industr Electron 67(3):2283–2293. https://doi.org/10.1109/TIE.2019.2907440
7. Xi X, Chen M, Zhang H, Zhou D (2018) An improved non-Markovian degradation model with long-term dependency and item-to-item uncertainty. Mech Syst Signal Process 105:467–480. https://doi.org/10.1016/j.ymssp.2017.12.017
8. Wang P, Youn BD, Hu C (2012) A generic probabilistic framework for structural health prognostics and uncertainty management. Mech Syst Signal Process Mech Syst Signal Process, 28,https://doi.org/10.1016/j.ymssp.2011.10.019
9. Zschech P, Bernien J, Heinrich K (2019) Towards a taxonomic benchmarking framework for predictive maintenance: the case of NASA's Turbofan Degradation
10. Harakeh A (2017) Adversarial Robustness of uncertainty aware deep neural networks
11. Kendal, A, Gal Y (2017) What uncertainties do we need in Bayesian deep learning for computer vision?
12. Prado A, Kausik R, Venkataramanan L (2019) Dual neural network architecture for determining epistemic and Aleatoric uncertainties

13. Gustafsson FK, Danelljan M, Schon TB (2020) Evaluating scalable Bayesian deep learning methods for robust computer vision. 2020 IEEE/CVF conference on computer vision and pattern recognition workshops (CVPRW), Seattle, WA, USA, 2020, pp 1289–1298, https://doi.org/10.1109/CVPRW50498.2020.00167
14. Le MT, Diehl F, Brunner T, Knol A (2018) Uncertainty estimation for deep neural object detectors in safety-critical applications. 2018 21st international conference on intelligent transportation systems (ITSC), Maui, HI, 2018, pp 3873–3878, https://doi.org/10.1109/ITSC.2018.8569637
15. Ghoshal B, Tucker A, Sanghera B, Wong WL (2019) Estimating uncertainty in deep learning for reporting confidence to clinicians when segmenting Nuclei image data. 2019 IEEE 32nd international symposium on computer-based medical systems (CBMS), Cordoba, Spain, 2019, pp 318–324,https://doi.org/10.1109/CBMS.2019.00072
16. Eldesokey A, Felsberg M, Holmquist K, Persson M (2020) Uncertainty-aware CNNs for depth completion: uncertainty from beginning to end. 2020 IEEE/CVF conference on computer vision and pattern recognition (CVPR), Seattle, WA, USA, 2020, pp 12011–12020, https://doi.org/10.1109/CVPR42600.2020.01203
17. Li G, Yang L, Lee C, Wang X, Rong M, A Bayesian deep learning RUL framework integrating epistemic and Aleatoric uncertainties. In: IEEE transactions on industrial electronics, https://doi.org/10.1109/TIE.2020.3009593
18. Araújo T, Aresta G, Mendonça L, Penas S, Maia C, Carneiro A, Mendonça AM, Campilho A (2020) DR|GRADUATE: uncertainty-aware deep learning-based diabetic retinopathy grading in eye fundus images. Medical Image Anal 63:101715, ISSN 1361-8415,https://doi.org/10.1016/j.media.2020.101715
19. Qin Y, Liu Z, Liu C, Li Y, Zeng X, Ye C (2021) Super-resolved q-space deep learning with uncertainty quantification. Med Image Anal 67:101885, ISSN 1361-8415,https://doi.org/10.1016/j.media.2020.101885
20. Godefroy G, Arnal B, Bossy E (2021) Compensating for visibility artefacts in photoacoustic imaging with a deep learning approach providing prediction uncertainties. Photoacoustics 21:100218, ISSN 2213-5979,https://doi.org/10.1016/j.pacs.2020.100218
21. Chai Y, Bian Y, Liu H, Li J, Xu J (2021) Glaucoma diagnosis in the Chinese context: an uncertainty information-centric Bayesian deep learning model. Inform Process Manag 58(2):102454, ISSN 0306–4573, https://doi.org/10.1016/j.ipm.2020.102454
22. Liu Y, Qin H, Zhang Z, Yao L, Wang Y, Li J, Liu G, Zhou J (2019) Deriving reservoir operation rule based on Bayesian deep learning method considering multiple uncertainties. J Hydrol 579:124207, ISSN 0022-1694,https://doi.org/10.1016/j.jhydrol.2019.124207
23. Tanno R, Worrall DE, Kaden E, Ghosh A, Grussu F, Bizzi A, Sotiropoulos SN, Criminisi A, Alexander DC (2021) Uncertainty modelling in deep learning for safer neuroimage enhancement: demonstration in diffusion MRI. NeuroImage 225:117366, ISSN 1053-8119,https://doi.org/10.1016/j.neuroimage.2020.117366
24. Hernández S, López JL (2020) Uncertainty quantification for plant disease detection using Bayesian deep learning. Appl Soft Comput 96:106597, ISSN 1568-4946,https://doi.org/10.1016/j.asoc.2020.106597
25. Ruben M (2014) BayesOpt: a Bayesian optimization library for nonlinear optimization, experimental design and bandits. J Mach Learn Res 15(1):3735–3739
26. Sherstinsky A (2018) Fundamentals of recurrent neural network (RNN) and long short-term memory (LSTM) network
27. Zheng S, Ristovski K, Farahat A, Gupta C (2017) Long short-term memory network for remaining useful life estimation, 88–95. https://doi.org/10.1109/ICPHM.2017.7998311
28. Ge Y, Wu J, Jiang X (2019) A prediction method using bayesian theory for remaining useful life. 2019 international conference on quality, reliability, risk, maintenance, and safety engineering (QR2MSE), Zhangjiajie, China, pp 856–862, https://doi.org/10.1109/QR2MSE46217.2019.9021252
29. Ramasso E, Saxena A (2014) Performance benchmarking and analysis of prognostic methods for CMAPSS datasets. Int J Prognost Health Manag 5:1–15

30. Saxena A, Goebel K, Simon D, Eklund N (2008) Damage propagation modelling for aircraft engine run-to-failure simulation. International conference on prognostics and health management.https://doi.org/10.1109/PHM.2008.4711414.M
31. Gao Y, Zhou J, Wu K, Zhao G, Hu C (2019) Construction method of turbine engine health indicator based on deep learning. 2019 prognostics and system health management conference (PHM-Qingdao), Qingdao, China, pp 1–6, https://doi.org/10.1109/PHM-Qingdao46334.2019.8943055
32. Wang Q, Zheng S, Farahat A, Serita S, Gupta C (2019) Remaining useful life estimation using functional data analysis. 2019 IEEE international conference on prognostics and health management (ICPHM), San Francisco, CA, USA, 2019, pp 1–8, https://doi.org/10.1109/ICPHM.2019.8819420
33. Li J, Li X, He D (2019) Domain adaptation remaining useful life prediction method based on AdaBN-DCNN. 2019 prognostics and system health management conference (PHM-Qingdao), Qingdao, China, pp 1-6,https://doi.org/10.1109/PHM-Qingdao46334.2019.8942857
34. "Detect small changes in mean using cumulative sum" MATLAB MathWorks® https://www.mathworks.com/help/signal/ref/cusum.html
35. Heimes F (2008) Recurrent neural networks for remaining useful life estimation. In: IEEE international conference on prognostics and health management
36. Bruneo D, De Vita F (2019) On the use of LSTM networks for predictive maintenance in smart industries. 2019 IEEE international conference on smart computing (SMARTCOMP), Washington, DC, USA, pp 241–248, https://doi.org/10.1109/SMARTCOMP.2019.00059
37. Babu GS, Zhao P, Li XL (2016) Deep convolutional neural network-based regression approach for estimation of remaining useful life. 21th international conference on database systems for advanced applications (DASFAA). Dallas, USA: Spinger, pp 214–228
38. Peng K, Pi Y, Jiao R, Dong J, Zhang K, Zhang C (2019) Remaining useful life prediction for aircraft engines based on grey model. 2019 prognostics and system health management conference (PHM-Qingdao), Qingdao, China, pp. 1–5,https://doi.org/10.1109/PHM-Qingdao46334.2019.8943000